ShopFlow

Solution Overview & Architecture

Quick Summary (30-second read)

What: Real-world order-processing system demonstrating SOLID principles and Clean Architecture in Python
Why: Bridges software engineering theory and production practice for Python developers
How: Practical implementation of SOLID, Clean Architecture, and testable design patterns
Status: Educational reference implementation, companion to published book
Book: "SOLID Principles in Python: A Hands-On Guide for All Developers"
GitHub: github.com/python-projects-fernando/shopflow

Continue reading for more details and methodology.

Key Takeaways for Decision Makers

✓ **Practical reference:** Real-world example, not abstract theory or toy code
✓ **Educational value:** Companion to published book with clear explanations
✓ **Production patterns:** Clean Architecture, dependency inversion, interface segregation
✓ **Fully testable:** Comprehensive pytest suite with isolated unit tests
✓ **Transparent design:** Both correct implementations and common violations documented

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Value Proposition

Educational reference implementation that demonstrates how SOLID principles and Clean Architecture translate into clean, maintainable, testable Python code.

Built for **Python developers, engineering teams, and technical leaders** who want to move beyond tutorial hell and understand how design principles apply in practice.

Key Benefits:

- Realistic order-processing system modeling e-commerce backend workflows

- Clear demonstration of SOLID principles with before/after comparisons
- Clean Architecture layering (domain, application, infrastructure)
- Full testability with pytest and isolated unit tests (no real DB or email)
- Explicit dependency inversion and interface segregation patterns

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Business Impact & Expected Outcomes

Measurable Value for Your Organization

Metric	Target Impact	Why It Matters
Code Maintainability	Improved structure	Clean Architecture reduces technical debt and simplifies future changes
Developer Onboarding	Faster ramp-up	Clear patterns and documentation accelerate team understanding
Test Coverage	Comprehensive unit tests	Isolated tests enable confident refactoring and regression prevention
Design Quality	SOLID adherence	Reduces coupling and improves long-term code evolution

Core Features That Deliver Value

- Order creation with inventory validation
- Payment processing with explicit dependency injection
- SMS and email notifications via interface segregation
- Customer and product domain entities with clear boundaries
- SQLite database with repository pattern abstraction
- Comprehensive pytest test suite with isolated units

Our Approach: Educational Reference Implementation

Why This Matters for Your Project

ShopFlow is an **educational reference implementation** demonstrating how SOLID principles, Clean Architecture, and modern Python practices deliver real business value through maintainable, testable code.

Result: When you engage for implementation or consulting, you receive guidance based on proven patterns, reducing risk, rework, and long-term maintenance cost.

Core Design Principles

Principle	Business Benefit
SOLID Principles	Reduces coupling, improves testability and maintainability
Clean Architecture	Core logic independent of frameworks and databases
Dependency Inversion	Easy to swap implementations without rewriting core logic

Principle	Business Benefit
Interface Segregation	Focused contracts reduce unnecessary dependencies
Test-Driven Design	Comprehensive tests enable confident refactoring

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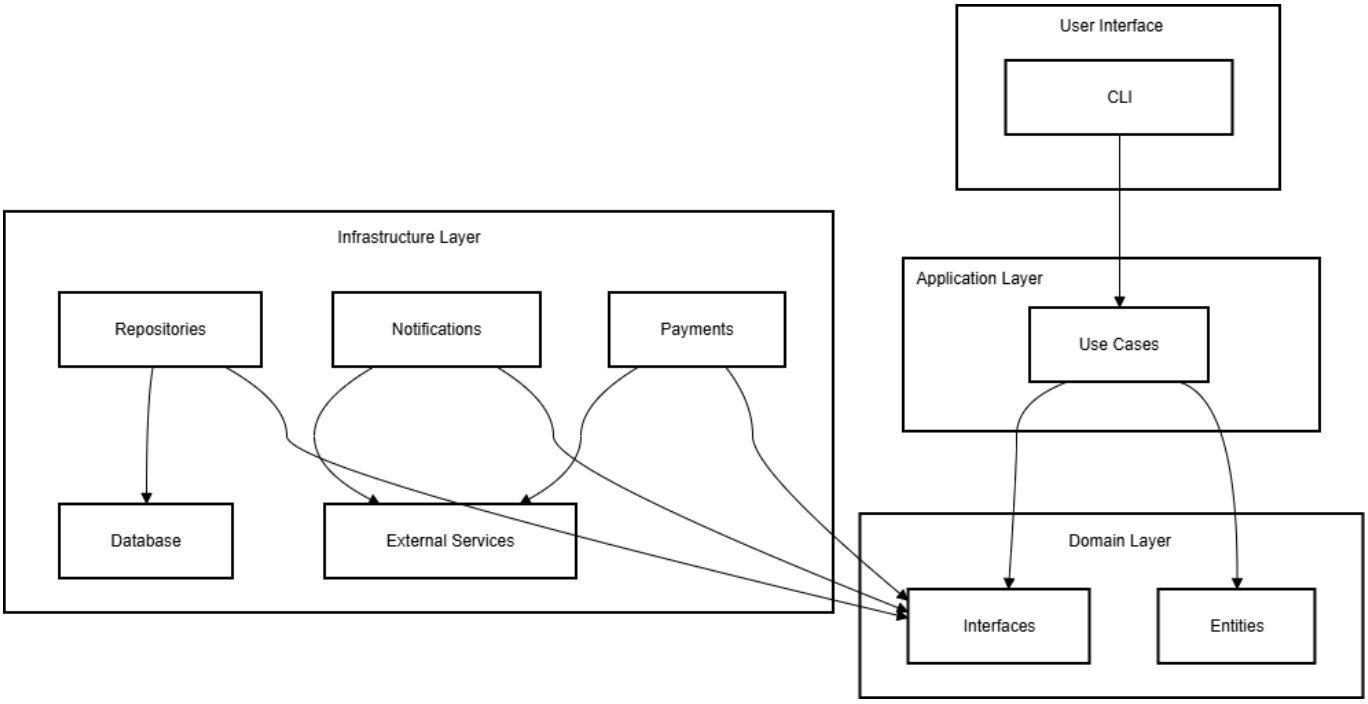
Solution Architecture Overview

High-Level Design



Why This Architecture Delivers Value

Clean Architecture was implemented because it:

- **Reduces long-term cost:** Core business logic independent of frameworks and databases
- **Accelerates testing:** Isolated domain rules enable reliable automated testing
- **Supports education:** Clear boundaries simplify teaching design principles
- **Future-proofs investment:** Easy to swap infrastructure without rewriting core logic

For technical readers: Detailed code examples and explanations available in the published book.

Technology Stack (Educational Reference)

Category	Technology	Rationale
Language	Python 3.11+	Strong typing, rich ecosystem, enterprise adoption
Architecture	Clean Architecture	Separation of concerns (domain, application, infrastructure)

Category	Technology	Rationale
Database	SQLite	Lightweight, no external dependencies, ideal for learning
Testing	pytest	Comprehensive unit and integration test coverage
Design Patterns	SOLID, Dependency Injection, Repository Pattern	Industry-standard patterns for maintainable code

All choices documented in the published book "SOLID Principles in Python".

Project Status

Current Phase: Educational Reference Implementation
Maturity: Functional, complete order-processing workflow implemented and tested
Transparency: Full codebase available on GitHub with book companion

What This Means for You:
This is not a production SaaS product, it's an educational reference implementation demonstrating production-grade patterns for Python development. When you engage for consulting or implementation, you receive guidance based on validated, documented patterns, not theory.

Get in Touch

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Book: Available on Amazon

Technical Appendix (Optional Deep-Dive)

For technical stakeholders who want implementation details.

Clean Architecture: Component Breakdown

Core Layers:

- **Domain:** Core business entities and rules (Order, Product, Customer, Payment)
- **Application:** Business logic orchestrators (Use Cases) and service interfaces
- **Infrastructure:** Database (SQLite), notification services (SMS/Email), repositories

Domain Entities:

- **Order:** Order creation with inventory validation and payment processing
- **Product:** Product entity with stock management
- **Customer:** Customer entity with contact information
- **Payment:** Payment processing with explicit dependency injection

Key Principles Applied:

- Dependencies point inward (Dependency Inversion)
- Core logic has zero framework dependencies
- External concerns isolated for easy testing and replacement
- Domain model aligned with e-commerce business requirements

SOLID Principles Demonstrated

- **Single Responsibility:** Each class has one reason to change
- **Open/Closed:** Open for extension, closed for modification
- **Liskov Substitution:** Subtypes can replace base types without breaking behavior
- **Interface Segregation:** Focused interfaces reduce unnecessary dependencies
- **Dependency Inversion:** Depend on abstractions, not concretions

Testing Strategy

- **Unit Tests:** pytest for domain logic and use cases
- **Isolation:** No real database or email dependencies in tests
- **Fast Execution:** All tests run quickly for immediate feedback
- **Comprehensive Coverage:** Core business logic fully tested

How to Run

Activate virtual environment:

```
python -m venv .venv
source .venv/bin/activate  # macOS/Linux
```

Install dependencies:

```
pip install -r requirements.txt
```

Run the application:

```
python main.py
```

Run the tests:

```
python -m pytest tests/
```

All components follow SOLID principles and Clean Architecture patterns.

This document reflects the current state of the ShopFlow educational reference implementation. All patterns and decisions are documented in the published book "SOLID Principles in Python".